

Basic Security Concepts

1. What is Information Security?

Information Security (InfoSec) means **protecting information** and **information systems** from unauthorized access, use, disclosure, modification, or destruction. Its main goal is to **ensure that information remains confidential, accurate, and available** to authorized users.

2. Goals of Information Security (CIA Triad)

The **CIA Triad** is the foundation of information security.

a. Confidentiality

- Ensures that information is only accessible to authorized persons.
- Protects data from unauthorized access or disclosure.
- **Examples:**
 - Using passwords and encryption.

b. Integrity

- Ensures that data remains **accurate and unaltered**.
- Prevents unauthorized modification or deletion.
- **Examples:**
 - Digital signatures.

c. Availability

- Ensures that systems and data are **available whenever needed**.
- Protects against downtime or service interruption.
- **Examples:**
 - Backups and disaster recovery plans.

3. Security Attacks

A **security attack** is any attempt to violate the CIA triad.

a. Passive Attacks

- Goal: To **observe or monitor** data without changing it.
- **Examples:**
 - Traffic analysis.

b. Active Attacks

- Goal: To **alter, damage, or disrupt** data or services.
- **Examples:**
 - Data modification.

4. Layers of Security

- **Physical Security:** Protects hardware and facilities.
 - *Example:* Locks, CCTV, access cards.
- **Network Security:** Protects network from intruders.
 - *Example:* Firewalls, VPN.
- **Application Security:** Protects software applications.
 - *Example:* Secure coding, input validation.
- **Data Security:** Protects stored and transmitted data.
 - *Example:* Encryption, masking.
- **User Security:** Involves user awareness and strong passwords.